

Portfolio Exam 2

Sensor Fusion – Particle Filter

Due date: 26.06.2026, 23:59

The deliverable for the following assignment is the commented source code of your implementation. Upload it via the eLearning platform for your group. The code should be structured in such a way that it can be executed without modification. So please include a readme file.

We will discuss your solution in a dedicated interview. During this, you should be able to present and explain your solutions, visualize results graphically, and change parameters. You do not need to prepare slides.

Task P2.1

Realize an implementation of the Particle Filter in a programming language of your choice for a simulation of the ball-throwing example from the lecture slides. The task of your Particle Filter is to estimate the positions and velocity vectors of $n \geq 1$ balls flying simultaneously only from the observed erroneous positions over time.

- Simulate the trajectory of n balls with the parameters *launch position* (x, y) (the imaginary ground is at $y = 0$), *launch speed* and *launch angle/launch direction* of each ball.

The initial positions and flying directions of the balls are unknown to your estimation. You can only assume that the ball starts within a rather large range of, for example, 50×50 meters.

- Simulate the observations of the ball positions (x, y) . The estimated ball positions shall be subject to high uncertainty and it shall be possible to parameterize this uncertainty. In addition, the time span between two observations shall be variable and observations may drop out completely over a certain period of time. It is necessary that the positions of the balls are also estimated during the time of the sensor failure.
- How do you deal with more than one ball flying at the same time? How do you define your state? Think intensively about what the transition model and the evaluation model should look like in the case of n balls.

How do you estimate n positions from the sensor fusion density? If the density of the ball positions is multimodal, select a suitable method that can determine the positions of the balls.

- You are not able to distinguish between the balls. They are indistinguishable from an observational point of view.
- The launch position and other launch parameters are just as unknown and cannot be specified more precisely. You should handle both the case where the launch positions and directions of the balls are very similar and the case where the positions and directions are clearly different in one common approach.

You must be able to visualize the underlying particle sets over time. Within this visualization, you can also show your estimates (e.g. ball positions, ball velocities) for the ball states.

Note: There is more than one good approach here. Thus, there is no clear best solution approach in this task.